LADON DAVIS

Software Engineer — Backend & Full-Stack

Seattle, WA · US work authorization · cjdavis766@gmail.com · viceatlas.com · github.com/viceatlas

Building and shipping software since 2020, alone. I build systems that grade their own work: a live consumer product on Next.js, TypeScript and Postgres; Python decision engines with an append-only audit ledger; LLM pipelines on the Anthropic SDK that score their own output.

SELECTED PUBLIC WORK

- **self-grading-demo** — agent pipeline that grades its own output: LLM-as-judge, sha256-frozen rubric, hash-chained ledger; runs keyless in under a second. github.com/viceatlas/self-grading-demo
- **role-scout** — job-discovery pipeline over ATS APIs; the model scores, a human approves every action. github.com/viceatlas/role-scout
- **viceatlas.com** — the live product below; predictions locked before events, graded in public; health at /api/health. viceatlas.com

EXPERIENCE

Software Engineer — Independent (Self-employed) | **2020 – Present** | **Seattle, WA**

Sole engineer on a live consumer product, a family of quantitative decision engines, an LLM research pipeline and three client websites, from schema to deploy to the incident write-ups.

ViceAtlas — consumer web product, live at viceatlas.com

- Built and operate a production Next.js/TypeScript app on Supabase/Postgres and Vercel: **1,355 prerendered pages** (build of Aug 31, 2026), Discord OAuth and magic-link sign-in, 47 re-runnable migrations and 84 row-level-security policy statements (repo count, Oct 2026), and a **/api/health** operations endpoint.
- Hardened it for production: fail-open middleware, Postgres rate-limit triggers, a moderation pipeline that fails closed to a human review queue, a re-runnable live-fire certification against prod (20-route sweep, 8-stage account-lifecycle test with an RLS negative case), two secret-authenticated GitHub Actions cron workflows (100 of 100 recent runs green, Oct 2026), and a dated failure ledger placed on a severity ladder.

Quantitative decision engines — Python

- Shipped a risk-gated options decision engine (FastAPI, Pydantic v2, SQLAlchemy async) with an **enforced conservatism hierarchy**: a layer can only tighten the decision, and a runtime check fails the run on violation. Graded against the next open: 60 of 66 nightly decisions, Apr–Oct 2026; 33 traded nights, 2 short-strike breaches.
- Made the record tamper-evident: **write-once decision freezes** before each event, an **append-only cross-system SQLite ledger** (two INSERT-only tables, nine write-time fences, 29 hard-fail checks), and an operations manual whose checker recomputes 96 probes against code, config and a read-only store and exits nonzero on drift.
- Built **walk-forward, leak-free calibration** with shrunk-Wilson lower bounds per bucket, and purged, embargoed walk-forward splits with block-bootstrap verdicts over **10.1M labeled outcomes** (9 years of market sessions, Jul 2026 run); 619 tests collected on the flagship (Oct 2026). Retired strategies on measured results, one against a kill bar frozen in the ledger before the data arrived.

LLM research pipeline — Anthropic SDK, on a VPS cron since July 2026

- Built a **seven-stage LLM pipeline** (plan, search, fetch, extract, synthesize, verify, package) in a TypeScript monorepo (Fastify, BullMQ, Prisma): streamed responses, strict JSON output contracts with fail-loud parsing, per-step token metering in Postgres; 411 test cases across 25 vitest files (static count, Oct 2026).
- Evaluation in the production path: an **adversarial LLM-as-judge** scores every claim against its fetched evidence under a versioned rubric (81 of 84 claims judged supported across the six production runs Aug 14–Oct 8, 2026, 95% CI 90–99%; not yet validated against human labels); a prompt-version calibration gate exits nonzero when a new prompt regresses the prior version's graded record.

SKILLS

Languages: TypeScript, Python, SQL, Node.js
Backend & data: Next.js (SSG/ISR), React, FastAPI, Pydantic v2, SQLAlchemy (async), REST APIs, OAuth, PostgreSQL (migrations, row-level security, triggers), Supabase, SQLite
Infra & ops: Linux VPS, cron orchestration, GitHub Actions, Vercel, health checks, fail-closed design, pytest, vitest, Playwright, Git
AI/LLM: Anthropic SDK, OpenAI API, multi-stage LLM workflows, strict JSON output contracts, LLM-as-judge evaluation